南京邮电大学2025-2026第二学期《软件工程（双语）》期末试卷（A）

1. True-False Question 2*5=10pts

1. At the inception of requirements engineering, the difference between project management configuration (SCM) and software support is equivalent.
2. A good software design should have high cohesion within modules and low coupling between modules.
3. Planning ahead for reuse reduces the cost and increases the value of both the reusable components and the systems into which they are incorporated.
4. Software is tested to verify the correctness of the system as it was designed and constructed.
5. User experience (UX) design is the process of enhancing user satisfaction with a product by creating a usable, accessible, and pleasurable interaction between product and its users.

2. Problem Analysis 6*5=30pts

1. What information is managed in software configuration management?
2. What is the difference between black-box testing and white-box testing?
3. Why should requirements engineering be an iterative process?
4. What are the questions that need to be answered using the $W5HH$?
5. How do requirement engineers collaborate with stakeholders?

3. Comprehension & Communication 10*3=30pts

3.1 Question 1

Consider the following two questions typically asked during the initial stages of a project:

Who will use the solution?

What problem(s) will this solution address?

Based on these questions, please provide a detailed analysis addressing the following:

3.1.1 The respective purpose of the two questions above.

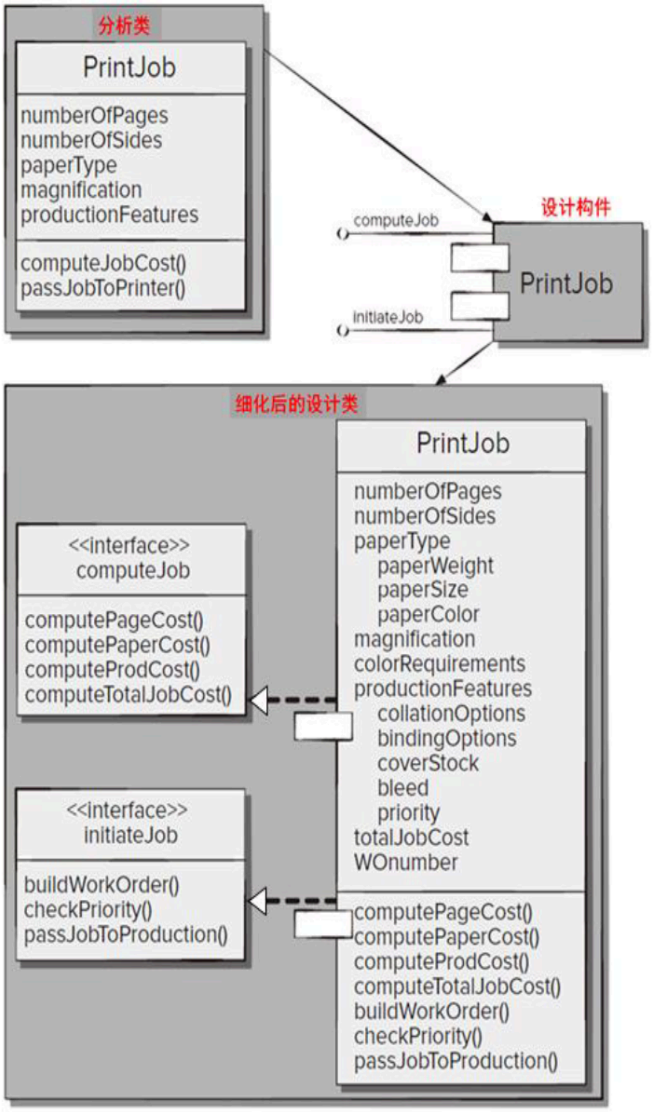
3.1.2 The respective impact of the two questions above on requirement gathering.

3.2 Question 2

Since the suppliers of products are not in the same city, when you place an order(订单) as shown in the figure on an e-commerce platform through the shopping cart(购物车), the system needs to provide the function of splitting the order into some delivery orders. Please model this function with UML swimlane(泳道) activity diagram.

序号	商品编号	商品名称	供应商地址
01	C01	华为手机	上海
02	S03	打印纸	北京
03	D01	滑板	广州
04	N01	牛仔裤	北京
05	C02	iPad	上海
……	……	……	……

3.3 Question 3

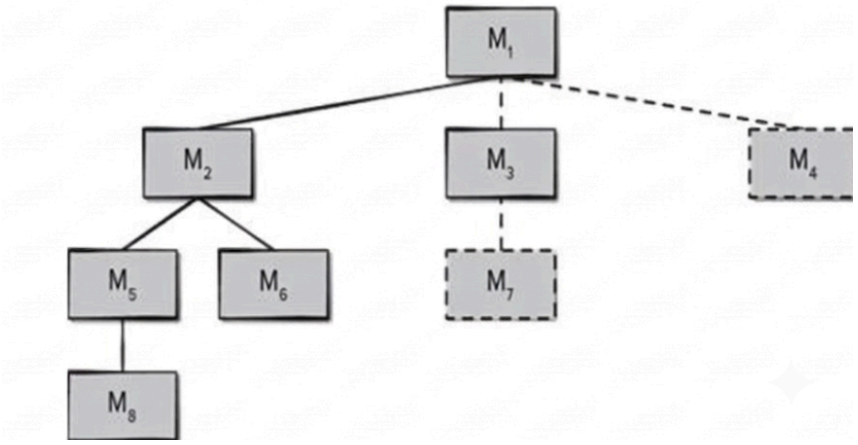


3.3.1 What are the flaws or issues with the current design?

3.3.2 How to refactore the next image?

4. Solution design/development 15*2=30pts

4.1 Question 1



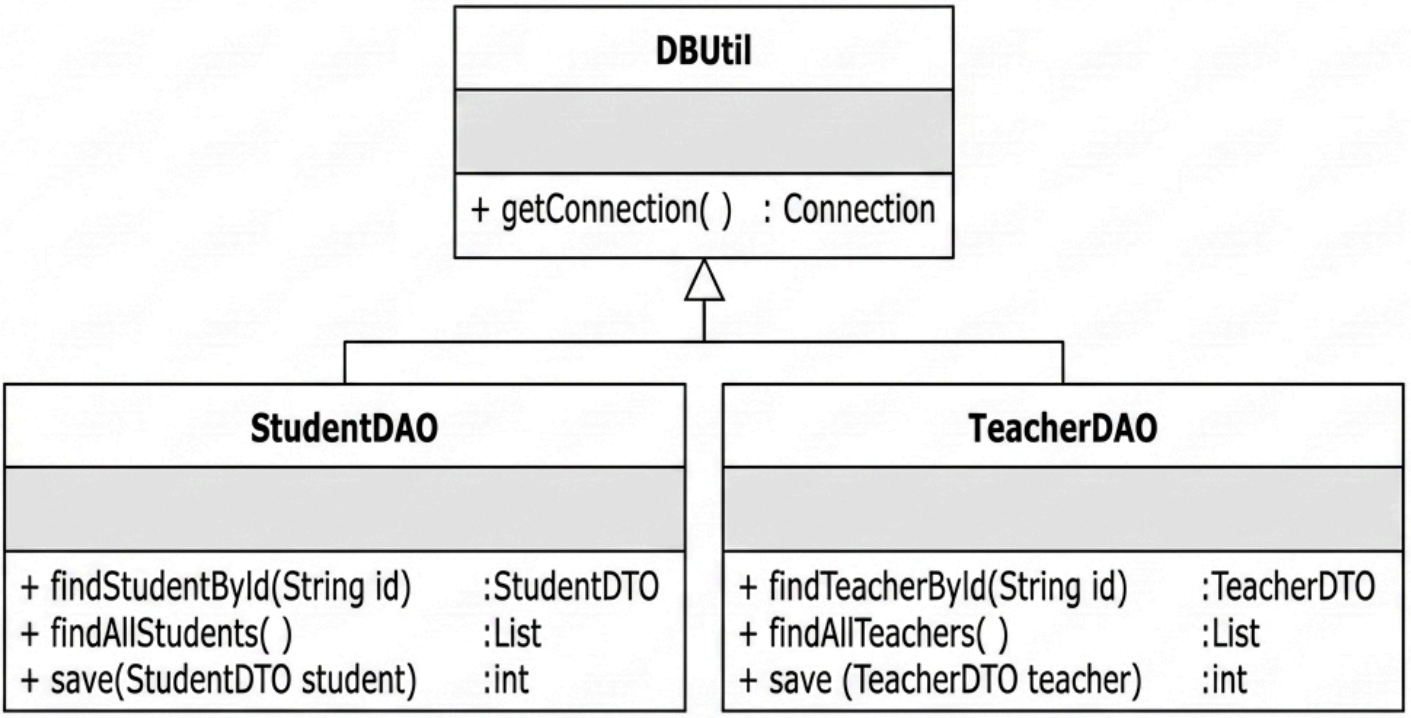
4.1.1 What is the difference between Depth-First Search (DFS) and Breadth-First Search (BFS)?

4.1.2 Based on the provided diagram, please provide the traversal sequences for both Depth-First Search (DFS) and Breadth-First Search (BFS).

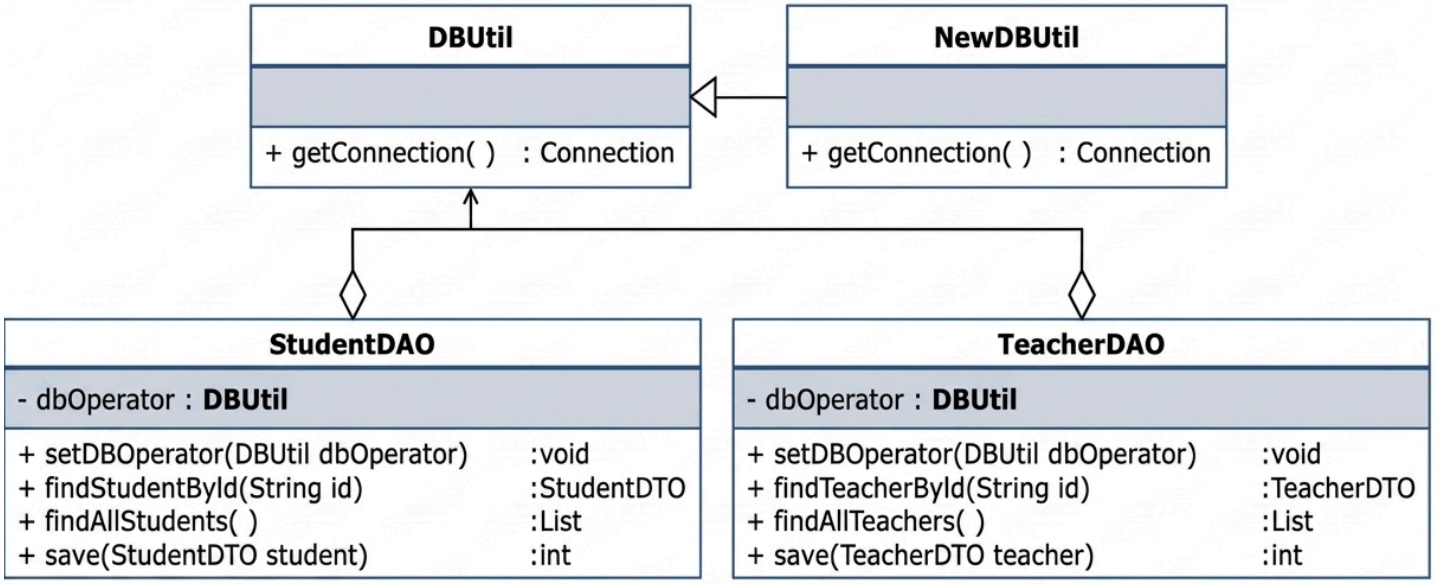
4.1.3 Which traversal method is typically preferred in large-scale software applications, and what is the reasoning behind this choice?

4.2 Question 2

For accessing the database in a certain information system, Tom and Jerry have provided two design solutions respectively as shown in following diagrams (a) and (b):



Solution(A)



Solution(B)

Please answer the following questions:

4.2.1 What is the difference between the two solutions (a) and (b) regarding the reuse of method `getConnection()` ?

4.2.2 Which solution is better? State your reasons.